Intro to ML

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Agenda

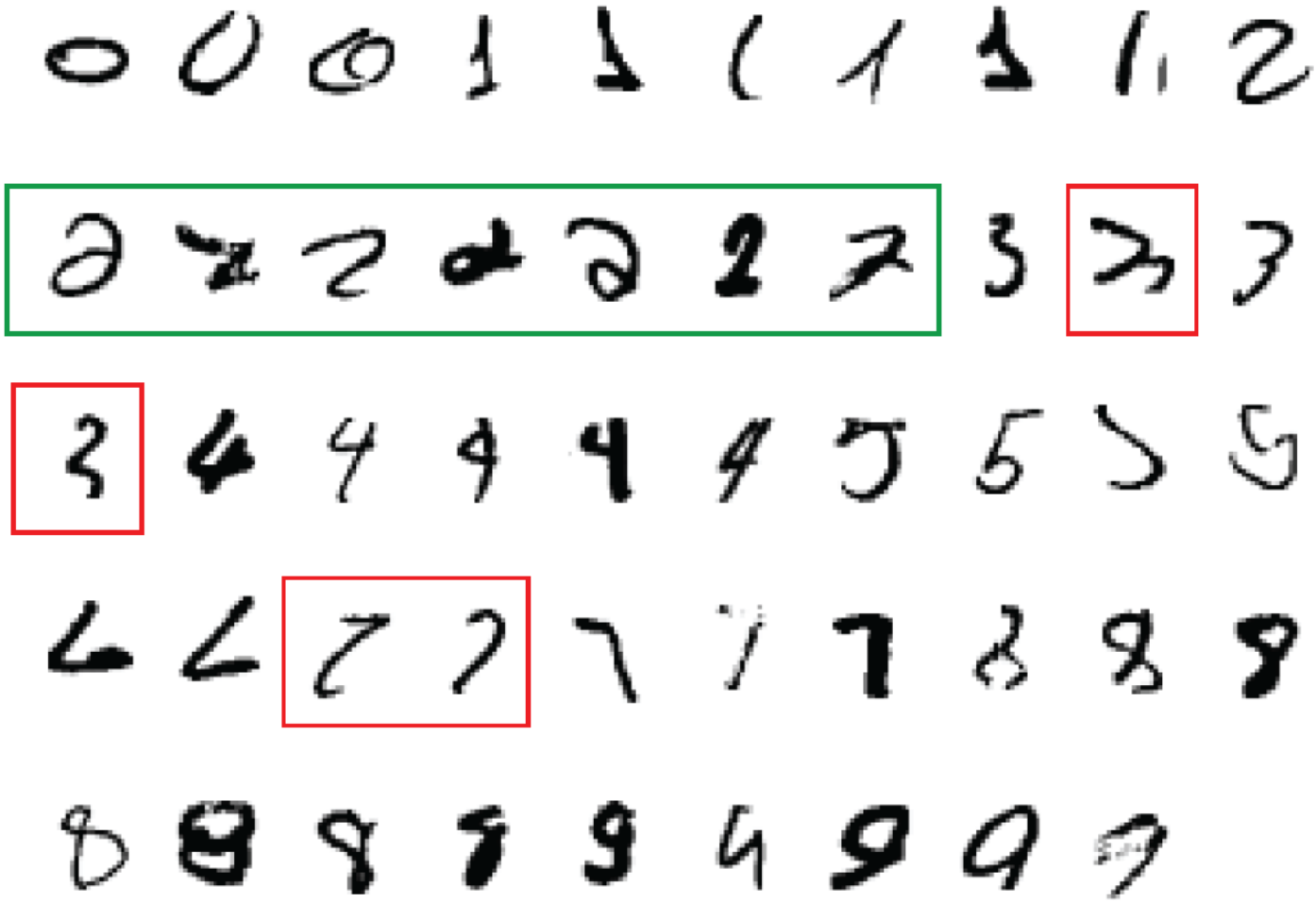
- Definition
- Examples of usage
- Ethics
- Terminology
  - Input
  - Output
  - Models

Definitions

- Machine Learning
  - Study of algorithms that
  - improve their performance
  - at some task
  - with experience
- Optimize a performance criterion using example data or past experience.
- Role of Statistics: Inference from a sample
- Role of Computer science: Efficient algorithms to
  - Solve the optimization problem
  - Representing and evaluating the model for inference

EXAMPLES OF USAGES

It is very hard to say what makes a 2



Learning Associations

- Basket analysis:

  $P(Y \mid X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

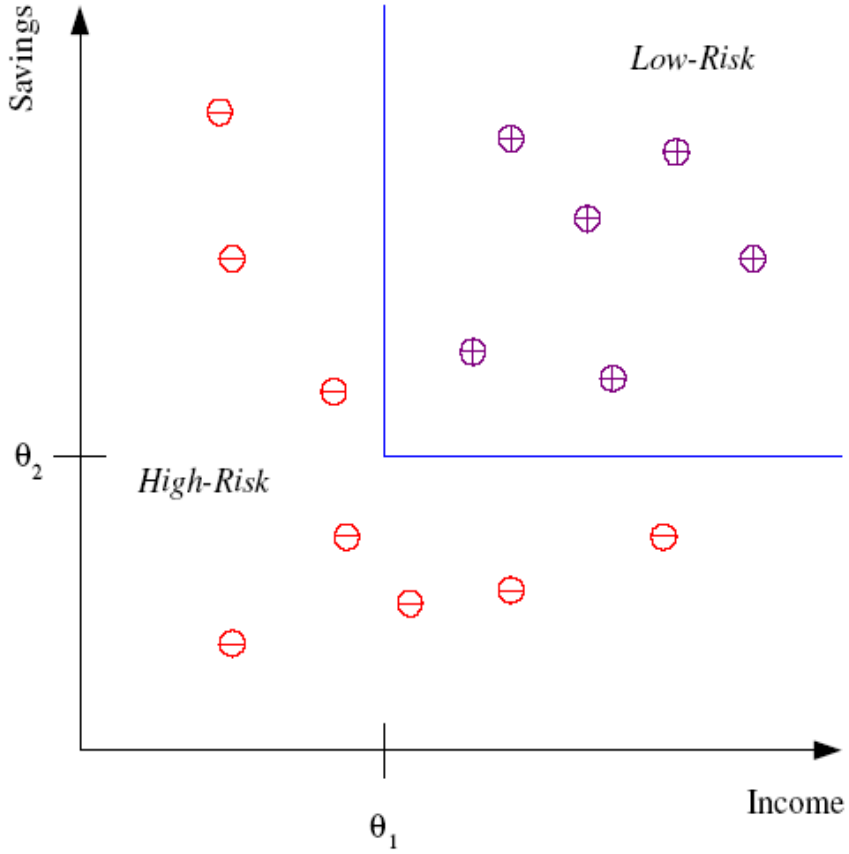
  Example: $P(\text{chips} \mid \text{beer}) = 0.7$

Market-Basket transactions

TID	*Items*
1	**Bread, Milk**
2	**Bread, Diaper, Beer, Eggs**
3	**Milk, Diaper, Beer, Coke**
4	**Bread, Milk, Diaper, Beer**
5	**Bread, Milk, Diaper, Coke**

Classification

- Example: Credit scoring
- Differentiating between low-risk and high-risk customers from their *income* and *savings*



Discriminant: IF *income* > θ_1 AND *savings* > θ_2
THEN low-risk ELSE high-risk

Model

Classification: Applications

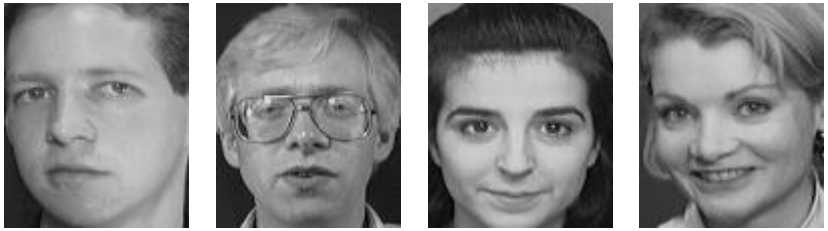
- Aka Pattern recognition
- Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style
- Character recognition: Different handwriting styles.
- Speech recognition: Temporal dependency.
  - Use of a dictionary or the syntax of the language.
  - Sensor fusion: Combine multiple modalities; eg, visual (lip image) and acoustic for speech
- Medical diagnosis: From symptoms to illnesses
- Web Advertizing: Predict if a user clicks on an ad on the Internet.

Face Recognition

Training examples of a person



Test images



AT&T Laboratories, Cambridge UK
http://www.uk.research.att.com/facedatabase.html

Prediction: Regression

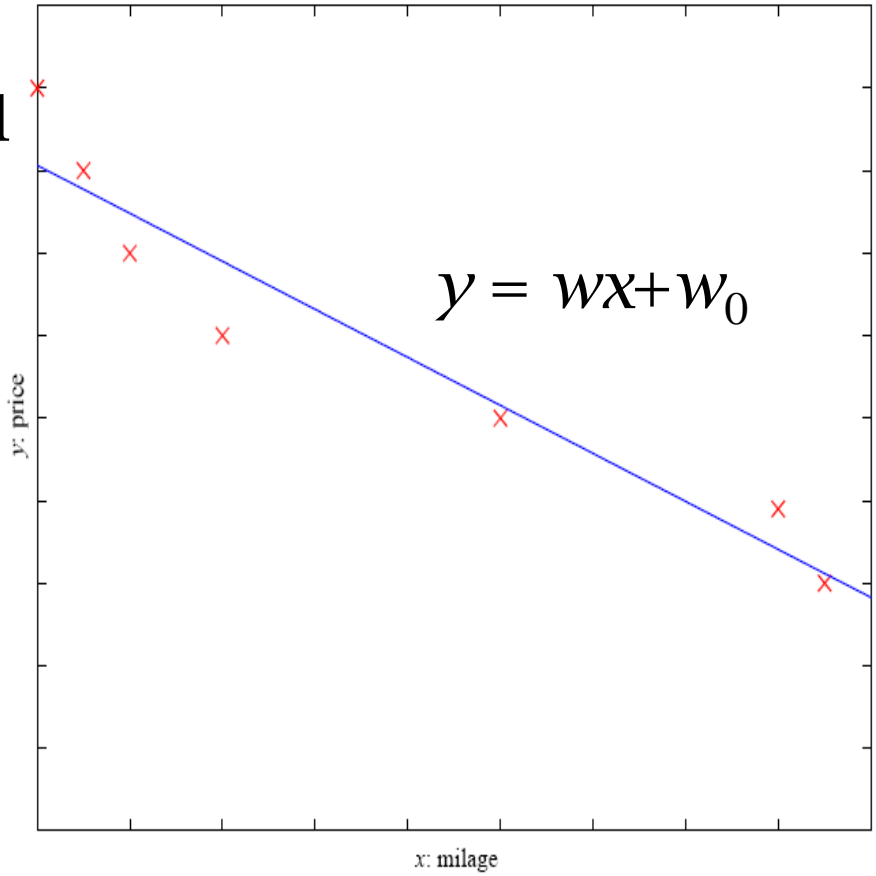
- Example: Price of a used car
- x : car attributes

  y : price

  $$y = g(x \mid \theta)$$

  $g(\,)$ model,

  θ parameters

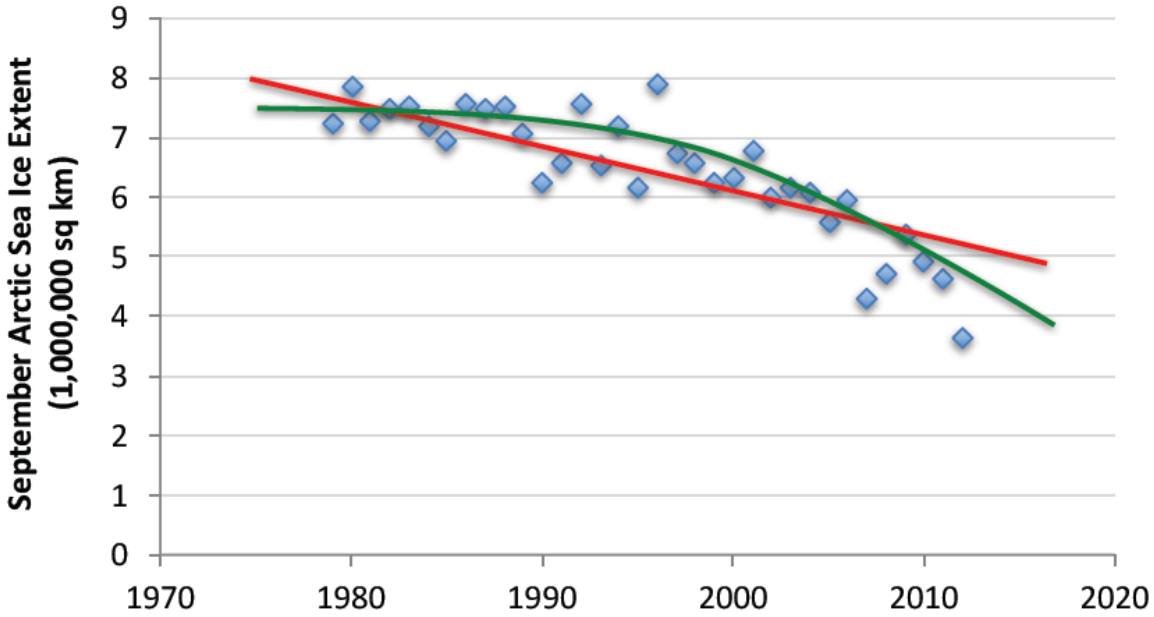


Types of learning

- **Supervised (inductive) learning**
  - Given: training data + desired outputs (labels)
- **Unsupervised learning**
  - Given: training data (without desired outputs)
- **Semi-supervised learning**
  - Given: training data + a few desired outputs
- **Reinforcement learning**
  - Rewards from sequence of actions

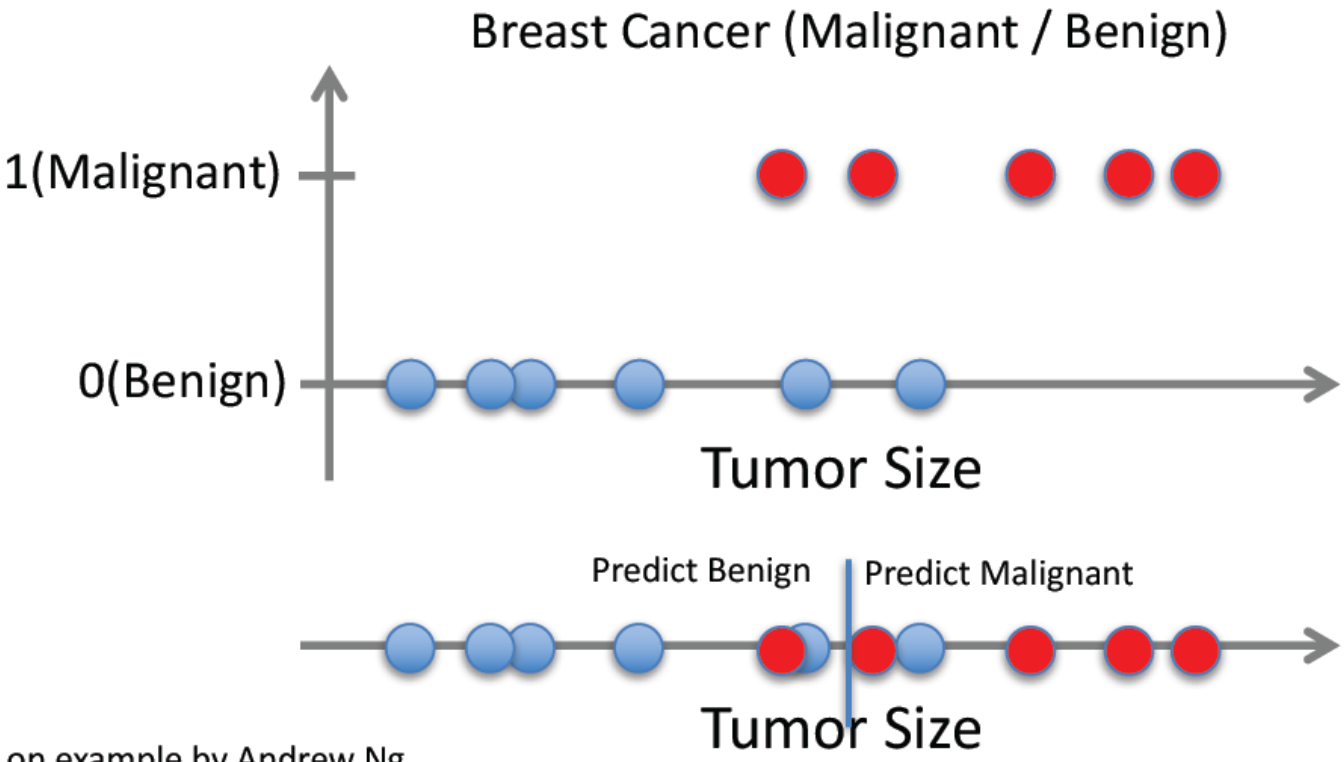
Supervised Learning: Regression

- Given (x_1, y_1), (x_2, y_2), …, (x_n, y_n)
- Learn a function $f(x)$ to predict y given x
  - y is real-valued == regression



Supervised Learning: Classification

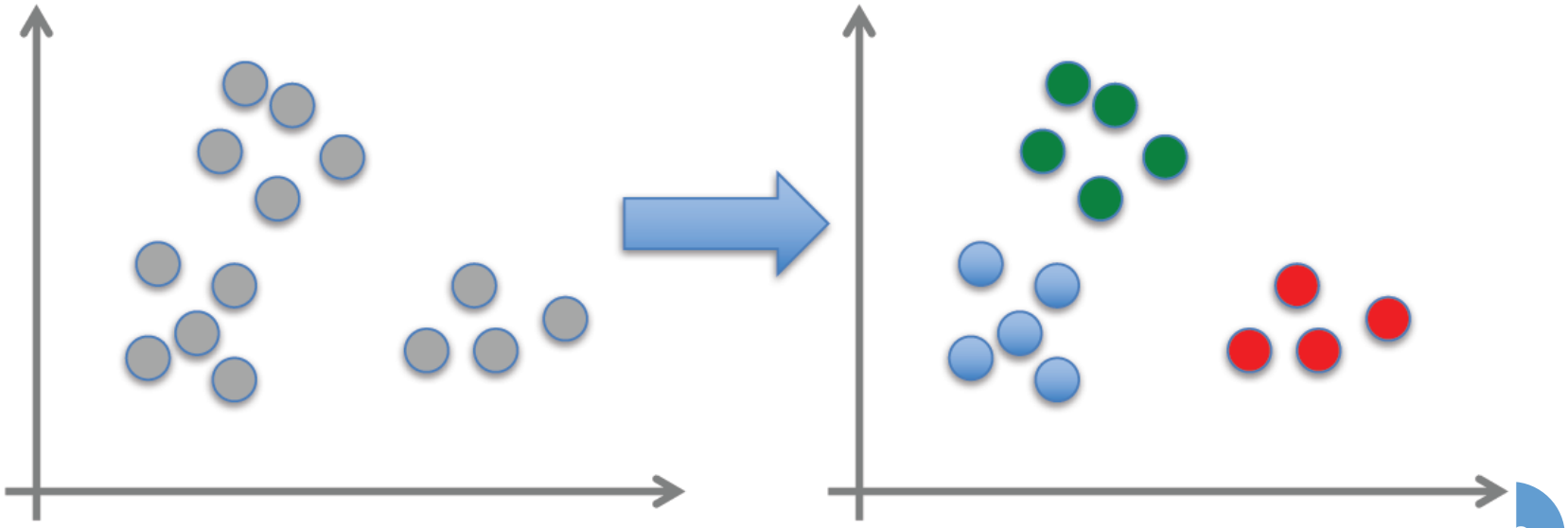
- Given (x_1, y_1), (x_2, y_2), ..., (x_n, y_n)
- Learn a function $f(x)$ to predict y given x
  - y is categorical == classification



Based on example by Andrew Ng

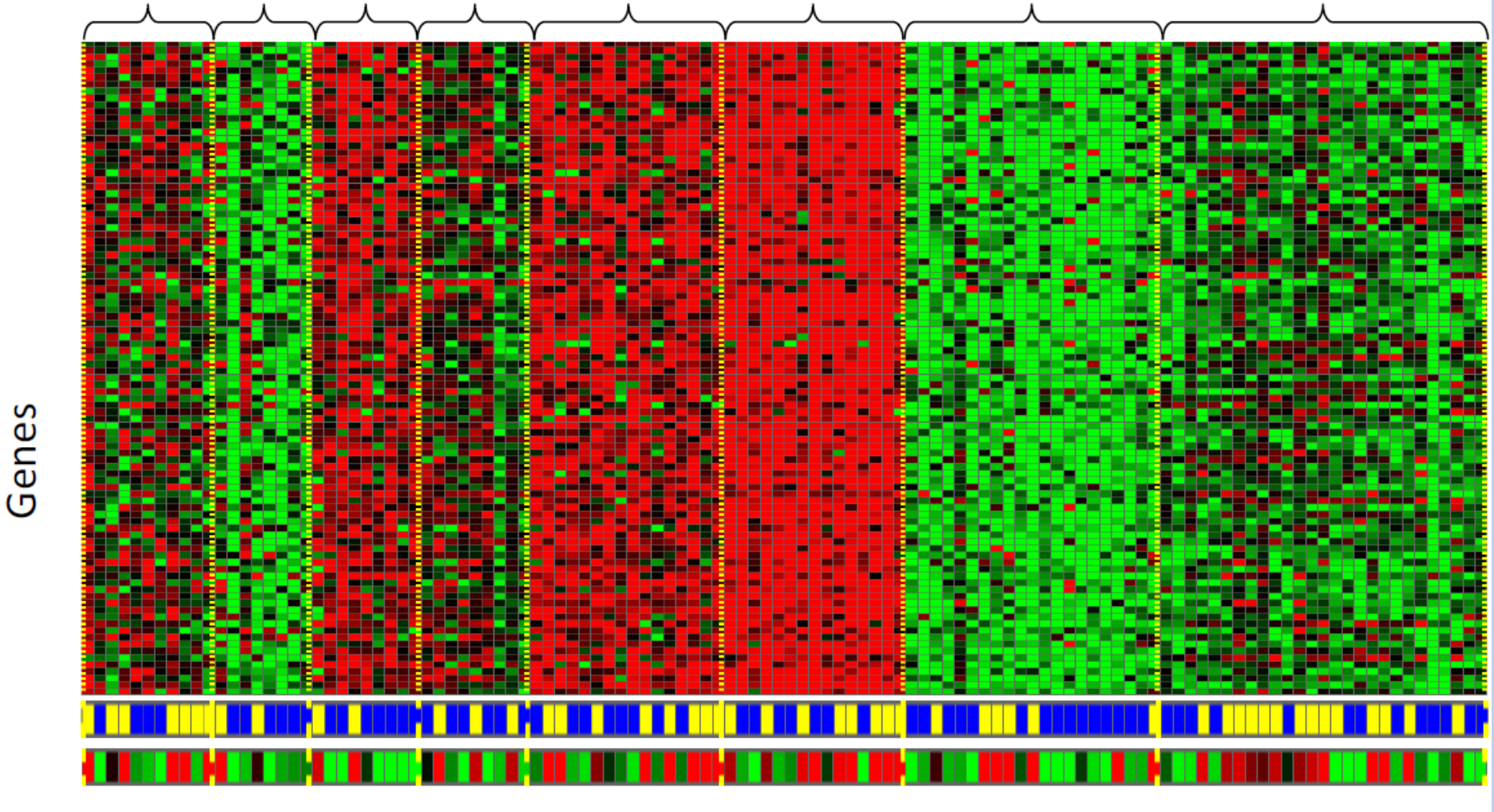
Unsupervised Learning

- Given x_1, x_2, …, x_n (without labels)
- Output hidden structure behind the x's
  - E.g., clustering



Unsupervised Learning

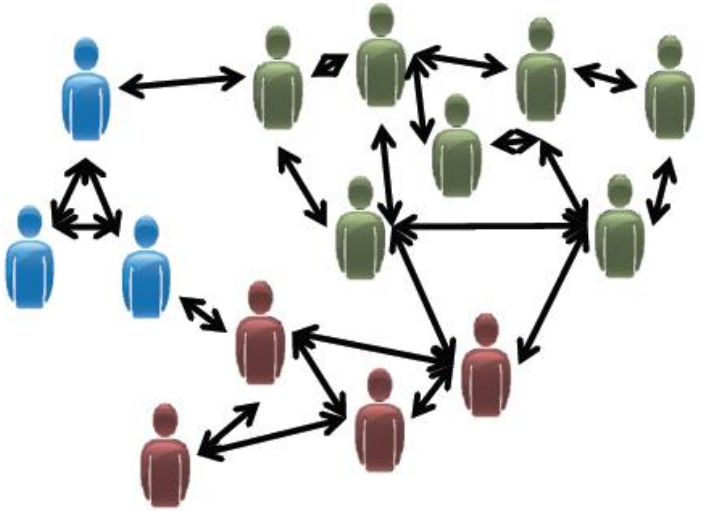
Genomics application: group individuals by genetic similarity



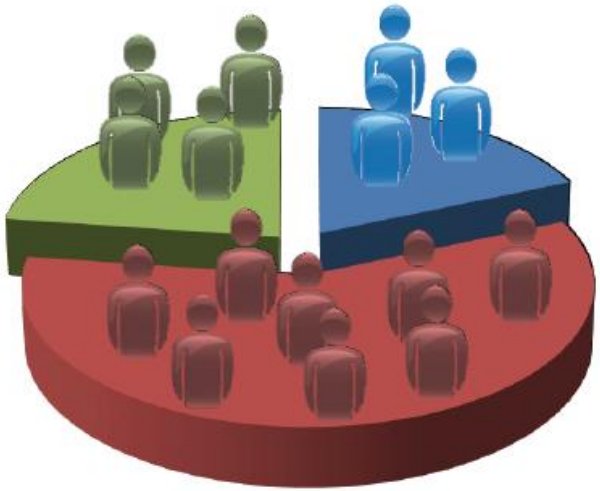
Unsupervised Learning



Organize computing clusters



Social network analysis



Market segmentation



Astronomical data analysis

Data mining and ethics I

- Ethical issues arise in practical applications
- Anonymizing data is difficult 85% of Americans can be identified from just zip code, birth date and sex
- Data mining often used to discriminate E.g., loan applications: using some information (e.g., sex, religion, race) is unethical
- Ethical situation depends on application E.g., same information ok in medical application
- Attributes may contain problematic information E.g., area code may correlate with race

Data mining and ethics II

- Important questions:
  - Who is permitted access to the data?
  - For what purpose was the data collected?
  - What kind of conclusions can be legitimately drawn from it?

Terminology: Input

- Instances: the individual, independent examples of a concept to be learned.
  - More complicated forms of input with dependencies between examples are possible.
  - Examples: Classes, humans, days, game day.
- Features/Attributes: measuring aspects of an instance.
  - We will focus on nominal and numeric ones.
  - Example: LOC, eye color, temperature, goals, etc.

Attributes

Instances

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	version	wmc	dit	noc	cbo	rfc	lcom	ca	ce	npm	lcom3	loc	dam	moa	mfa	cam	ic	cbm	amc	max_cc	avg_cc	Faultiness
2	1.3	11	4	2	14	42	29	2	12	5	0.725	395	1	1	0.885057	0.232323	3	4	34.54545	3	1.2727	FALSE
3	1.3	14	1	1	8	32	49	4	4	12	0.835165	257	1	0	0	0.307692	0	0	16.85714	6	1.6429	TRUE
4	1.3	3	2	0	1	9	0	0	1	1	0	58	1	1	0.714286	0.666667	1	1	17.33333	1	0.6667	FALSE
5	1.3	12	3	0	12	37	32	0	12	12	0.858586	310	1	1	0.770833	0.458333	0	0	24.08333	3	1.4167	FALSE
6	1.3	6	3	0	4	21	1	0	4	6	0.7	136	1	0	0.880952	0.416667	2	2	21	1	0.8333	FALSE
7	1.3	5	1	0	3	15	0	2	1	4	0.5	137	1	0	0	0.9	0	0	25.2	4	1.8	FALSE
8	1.3	4	4	0	6	32	6	0	6	1	2	325	0	0	0.963855	0.75	3	5	80.25	11	5.25	FALSE
9	1.3	16	3	0	3	40	84	1	2	16	0.658333	604	1	2	0.580645	0.291667	2	7	36.25	1	0.8125	FALSE
10	1.3	4	5	0	5	21	0	0	5	4	0.333333	87	1	0	0.980892	1	1	1	20.5	3	1.25	FALSE
11	1.3	17	3	0	10	55	76	5	6	11	0.59375	461	1	0	0.652174	0.285714	1	1	25.88235	6	2.6471	TRUE
12	1.3	9	3	0	9	29	0	0	9	9	0.583333	168	1	1	0.822222	0.444444	0	0	17.33333	2	1.4444	FALSE
13	1.3	3	3	0	3	14	1	0	3	3	0.5	84	1	0	0.948718	0.666667	1	1	26.66667	1	0.6667	TRUE
14	1.3	10	3	0	5	41	21	1	5	7	0.793651	360	1	0	0.804348	0.45	0	0	34.3	2	1.1	FALSE
15	1.3	17	2	0	9	64	76	7	2	9	0.879167	713	0.6	2	0.708333	0.339286	1	3	40.05882	9	1.4118	TRUE
16	1.3	5	1	5	12	11	8	11	1	5	0.75	59	1	0	0	0.7	0	0	10.6	5	1.6	FALSE
17	1.3	10	1	0	9	49	1	0	9	8	0.920635	382	1	0	0	0.833333	0	0	35.1	4	1.9	FALSE
18	1.3	22	4	0	16	72	165	1	16	16	0.904762	675	1	1	0.787879	0.227273	3	3	29.04545	4	1.2273	TRUE

Nominal attributes

- Values are distinct symbols
  - Values themselves serve only as labels or names
  - Nominal comes from the Latin word for name
- Values:
  - black, white, yellow, etc.
  - Binary: “defective / not defective”, (yes/no)
- Only equality tests can be performed

Numerical / Ratio quantities

- Ratio quantities are ones for which the measurement scheme defines a zero point
  - Example: Distance, number of bugs.
- All mathematical operations are allowed

Output: Knowledge representation

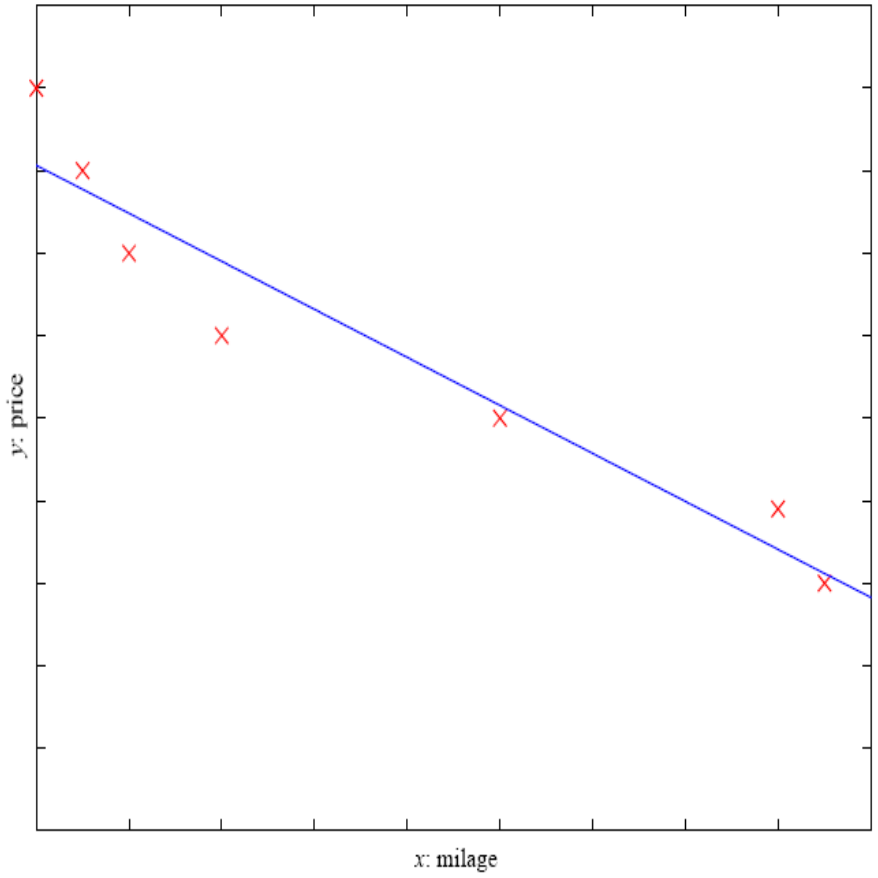
- Tables
- Linear models
- Trees
- Rules
- Classification rules
- Association rules
- Rules with exceptions
- More expressive rules
- Instance-based representation
- Clusters

Output: decision tables

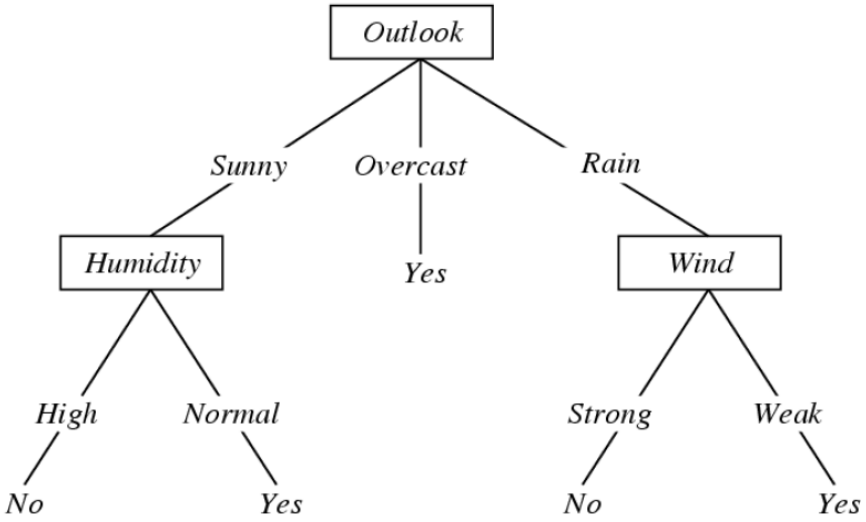
- Decision table for the weather problem:

Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Rainy	High	No
Rainy	Normal	No

Output: Linear models



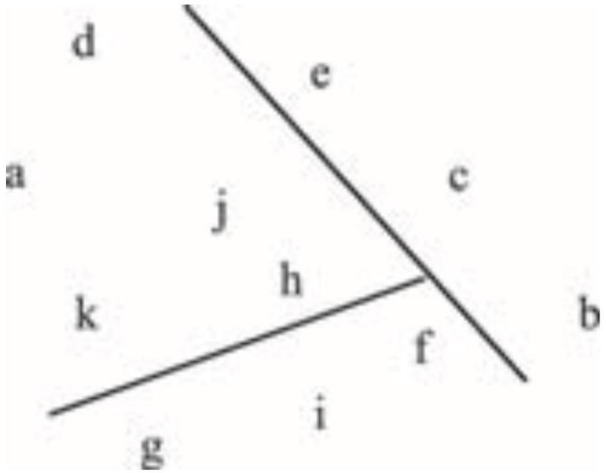
Output: Decision Tree



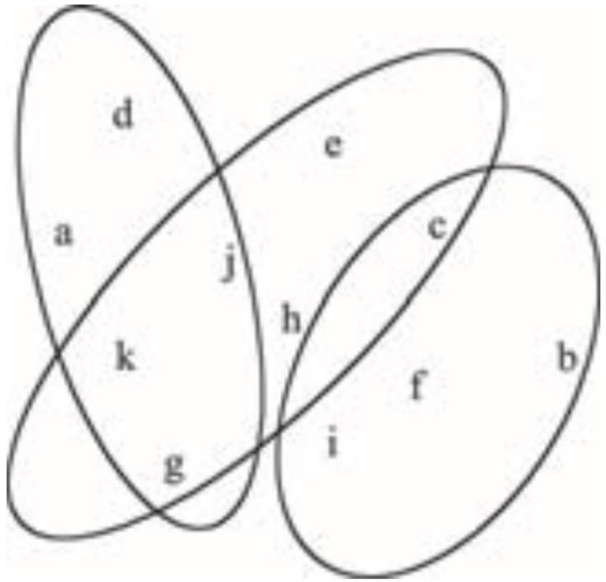
Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Output: Clusters

Simple 2-D representation



Venn diagram



Prediciton models

- Name
- Assumptions
- Type of input
- Type of output (classifiers / prediction models)

